Did a Teacher Platform Raise Grades? A Re-analysis of the ClassRoomMate Pilot

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Abstract

In the second semester of 2018, the learning-management platform ClassRoomMate was piloted in a public school in Santiago, Chile: 51 teachers, 27 classes from 4th grade to the last year of secondary school. Each teacher, together with the students of each class, chose how intensively to use it. The original evaluation compared the change of the class mean between the two semesters across usage levels and reported that intensive use raised performance by 3.5 % to 6.1 % and the probability of improvement to 75 %. We rebuilt the analysis from the school's grade books. The original numbers replicate exactly, but three of them overstate the evidence: the 6.1 % is an average over the improving classes only, the 75 % counts unchanged classes as improved, and the classes that chose intensive use started from a lower mean, so part of their gain is regression to the mean. Adjusted for the starting level and for the clustering of classes within teachers, intensive use is associated with a change of +1.2% (95% CI −1.3% to +3.6%), which is compatible with no effect as well as with a small positive one. With 13 to 15 intensive-use teachers, the pilot could only have detected differences of about 4 %. Because usage was self-selected and the outcome is the grades given by the same teachers, the data support an association at most, not an impact.

1 Context and question

ClassRoomMate is a platform for teachers: it records assessments, follows the progress of each class and gives feedback to students. In the second semester of 2018 it was piloted in a municipal (public) school in Santiago, Chile, in all classes from 4th grade of primary school (*básica*) to the 4th year of secondary school (*media*). The pilot covered 27 classes and 329 class–subject combinations (152 in primary, 177 in secondary school), taught by 51 teachers; religion, orientation and remedial lessons were excluded.

A first evaluation (2019, revised in 2025) compared the change of the class mean between the first and the second semester across three usage levels (*intensive*, *medium*, *low or none*) and concluded that intensive use had a positive and significant effect on performance. This report asks whether that conclusion holds:

1. Do the original numbers replicate from the original work sheet, and do they mean what the report says?
2. What remains of the association once the starting level of each class and the dependence between classes of the same teacher are taken into account?
3. What can such a pilot show at all, and how should the next one be designed?

Disclosure. The author collaborated with ClassRoomMate during the pilot. The data are the school's grade records; the school is not named, and no personal data are published (Section 2).

Table 1: Original summary by usage level, replicated from the work sheet. The original code kept the summary rows of the work sheet (block mean, maximum, minimum) as if they were classes; removing them changes little. $P(r \geq 1)$ is the original definition of improvement, $P(r > 1)$ excludes unchanged classes.

usage	classes	mean r	$P(r \geq 1)$ [%]	95% CI	$P(r > 1)$ [%]
original code (summary rows kept)					
intensive	116	1.022	74.1	[66.2, 82.1]	62.1
medium	70	1.001	62.9	[51.5, 74.2]	50.0
low / none	159	0.993	49.1	[41.3, 56.8]	39.6
summary rows removed					
intensive	110	1.023	74.6	[66.4, 82.7]	61.8
medium	64	1.003	64.1	[52.3, 75.8]	50.0
low / none	155	0.993	49.0	[41.2, 56.9]	39.4

2 Materials

Outcome. For every class–subject i (a class taught by one teacher), the ratio of the class means of the two semesters,

$$r_i = \bar{x}_{2,i} / \bar{x}_{1,i},$$

on the Chilean grade scale from 1 to 7. We analyse $\log r_i$, approximately the relative change; $r_i > 1$ means that the class mean rose.

Usage. The usage level was decided per teacher and class, by the teacher together with the students of that class. It is therefore self-selected at the level of the class.

Sources. The original evaluation worked from a work sheet holding one r_i per class–subject, grouped in six blocks (primary / secondary × three usage levels), but without the class each row belongs to. We recomputed the class means from the school's grade books (one export per class with every student's semester averages), linked each class–subject to its teacher through the staff list, and replaced the teachers' names by anonymous identifiers. Each work-sheet row was then located in its class: first by matching its ratio to a recomputed one (same level and subject; 116 rows), keeping only matches consistent with the class order of the work sheet, then by filling gaps between two matches of the same class (62 rows). The analysis uses the 142 located rows with known teacher and baseline: 47 teachers, 24 classes. Low-use classes are somewhat over-represented among them (56 % against 47 %), but the mean ratio of each usage level is the same as in the full work sheet (Tables 1 and 2).

Privacy. The grade books contain the names and grades of students, most of them minors, and the staff list contains the teachers' names. Neither is published. The code runs on the school's files and writes only class means with anonymous teacher identifiers, which stay private as well; the tests use synthetic grade books.

3 Replication of the original analysis

The original code reproduces the published numbers exactly (Table 1; logistic coefficient for intensive use 1.040, mixed-model coefficient 0.035, as in the original report). Four points change their reading.

1. **Summary rows counted as classes.** 16 summary rows of the work sheet (mean, maximum and minimum of each block) entered the analysis as classes (345 instead of 329 rows). The effect on the results is small.

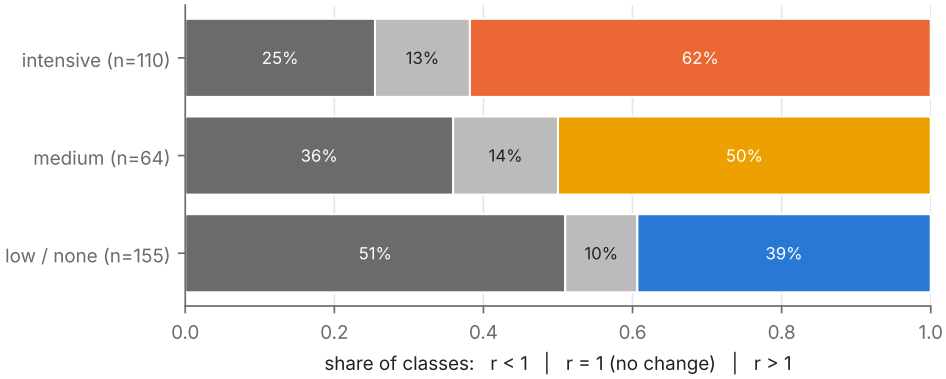


Figure 1: Share of classes whose mean fell, stayed the same (rounded to one decimal) or rose, by usage level (all 329 work-sheet rows).

Table 2: Located work-sheet rows used in the re-analysis, by usage level.

usage	classes	teachers	baseline (sem. 1)	mean r	$P(r > 1)$ [%]
intensive	42	15	5.63	1.022	69.0
medium	21	6	5.65	1.004	57.1
low / none	79	29	5.98	0.993	36.7

2. **No change counted as improvement.** Improvement was defined as $r \geq 1$. Because the class means were rounded to one decimal, 12 % of the classes have exactly $r = 1$. Counting only real improvements, the share is 62 % for intensive and 39 % for low use, instead of 75 and 49 % (Figure 1).

3. **The 6.1 % is conditional on improvement.** It is the mean increase over the intensive-use classes that improved. The same number for the low-use classes that improved is 4.3 %; over all intensive-use classes, the mean ratio is 1.023, against 0.993 for low use.

4. **Inconsistent tables.** The original tables report 110 intensive-use classes in one place and 116 in another, and the table of the medium usage level repeats the totals of the intensive one (110 classes, 41 improving, 37 %), whereas the work sheet has 64 classes with 41 improving (64 %).

Finally, both original models treat the classes as independent observations, or group them by subject. The decision on usage, however, was taken per teacher and class, so classes of the same teacher are not independent; and neither model accounts for the level at which a class started.

4 Re-analysis

Regression to the mean. Classes that start with a low mean tend to rise, and classes that start high tend to fall: one grade point more in the first semester goes with a change of −4.1% (95% CI −6.0% to −2.1%) in the second. The classes that chose intensive or medium use started 0.35 grade points lower than those with low use (Table 2 and Figure 2), so part of their gain would have occurred without the platform.

Models. Table 3 and Figure 3 estimate the same contrast under increasingly careful specifications. Without adjustment, intensive use goes with +2.7% ($p = 0.02$), close to the original estimate of the work sheet. Clustering the standard errors by teacher changes little here; adjusting for the starting level halves the estimate to +1.2% and its confidence interval includes zero,

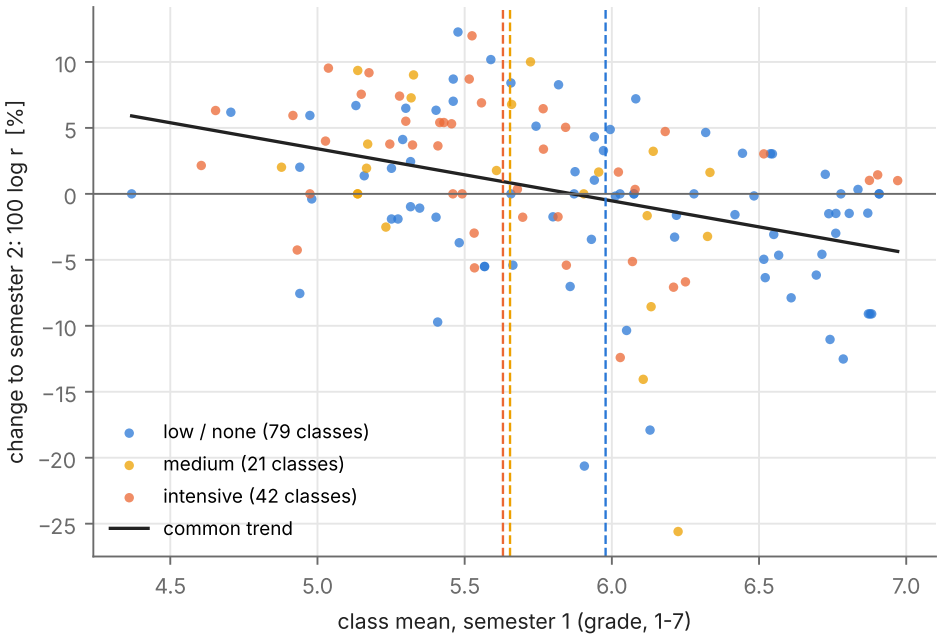


Figure 2: Change of the class mean against its level in the first semester (one point per class–subject). Dashed: mean starting level of each usage group. Intensive-use classes started lower and lie on the common trend.

Table 3: Difference in the semester change ($100 \log r$) with respect to low use. Baseline: class mean of the first semester. All specifications include primary/secondary level.

specification	intensive vs low		medium vs low	
	difference [%] (95% CI)	p	difference [%] (95% CI)	p
OLS, classes independent	+2.7 [+0.4, +5.1]	0.023	+0.7 [−2.4, +3.7]	0.656
OLS, SE clustered by teacher	+2.7 [+0.4, +5.1]	0.023	+0.7 [−3.0, +4.4]	0.714
+ baseline, SE clustered by teacher	+1.2 [−1.3, +3.6]	0.363	−0.9 [−4.9, +3.2]	0.674
+ baseline, SE clustered by class	+1.2 [−0.9, +3.3]	0.280	−0.9 [−4.9, +3.2]	0.673
mixed model: teacher + class, + baseline	+1.2 [−1.1, +3.4]	0.316	−0.9 [−3.7, +2.0]	0.551

whether the classes are clustered by teacher, by class, or modelled with random effects for both. Medium use shows no association in any specification.

Teachers as the unit. Usage was nearly constant per teacher (only 3 of 47 teachers have classes with different levels). Averaging the baseline-adjusted change per teacher and comparing the 13 intensive-use with the 28 low-use teachers gives a difference of +1.7%; a permutation test of the teachers' labels gives $p = 0.29$.

Power. With these teachers and the observed spread of their adjusted changes, the smallest difference that a two-sided test at the 5 % level would detect with 80 % power is about 4.2 %. The pilot was too small to detect effects much smaller than those it claimed.

5 What the pilot can and cannot show

Three features of the design limit any conclusion, independently of the statistics:

- **Self-selection.** Teachers and classes chose their usage level. Motivated teachers or classes may adopt the platform and improve for the same reason; the comparison of usage levels

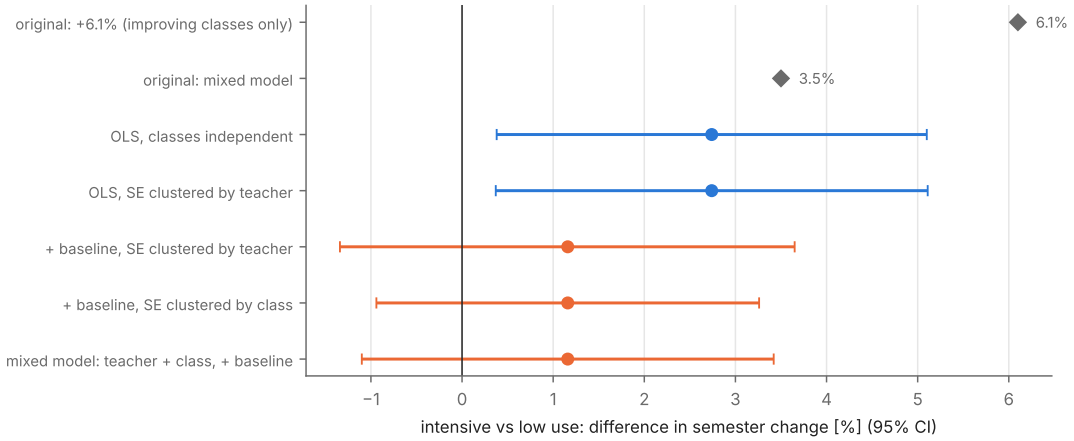


Figure 3: Intensive vs low use: the original figures (diamonds) and the estimates of Table 3 with 95% confidence intervals.

mixes the effect of the platform with the reasons for choosing it.

- **The outcome is set by the participants.** The grades are given by the same teachers whose usage is compared. A platform that structures assessment can change how grades are given, not only how much students learn.
- **No comparison over time or with another school.** Only one semester change is observed; there is no record of how the same classes changed in previous years.

A pilot that could show an effect would assign the platform at random to classes (or introduce it to groups of classes at random times), measure the outcome with assessments common to all classes and not set by the participating teachers (for example the national SIMCE tests or a common school-wide test), fix the analysis plan in advance, and include enough teachers: for a difference of 2 %, roughly four times as many as here.

6 Main changes

Table 4: Main changes with respect to the original evaluation.

	original evaluation	this version
Data	work sheet of ratios, without class or teacher	ratios recomputed from the grade books, linked to class and (anonymous) teacher
Improvement	$r \geq 1$	$\log r$; $r > 1$ when a share is reported
Summary rows	counted as classes	removed
Baseline	ignored	first-semester class mean as covariate
Dependence	classes independent or grouped by subject	clustered by teacher and class; teacher-level permutation test
Interpretation	“impact”, “effect”	association; power and design limits stated
Privacy	—	no personal data published; school not named
Report	notebook and LaTeX report	scripts for every table and figure, tests on synthetic grade books

7 Conclusions

1. The original numbers replicate exactly, but the headline figures overstate the evidence: the 6.1 % is an average over improving classes only, and the 75 % probability of improvement counts unchanged classes as improved.

2. Classes that chose intensive use started from a lower mean, and lower starting means go with larger gains. Accounting for this halves the raw association.

3. Adjusted for the starting level and the clustering of classes, intensive use goes with +1.2% (95% CI −1.3% to +3.6%): compatible with no effect and with a small positive one. Medium use shows no association.

4. The pilot could detect only differences of about 4 %, and its design (self-selected usage, grades given by the participating teachers, one semester) does not allow causal conclusions.

5. A randomised or staggered introduction with a common, external assessment would answer the question that this pilot could not.

Code

The code is available at `https://github.com/JuanCaChavaMor/classroommate-pilot`. The school's data are not included.